

Movie Shot Scale

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Outline

- Quantitative descriptive analysis
 - Data and missing values
 - Univariate summaries
 - Correlations
 - Summaries by country
- Surprising observations
 - Impossible ASL values
 - A genuine extreme: Rope
 - Extreme shot counts
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 - Correlation matrix
 - Pairs
 - Boxplots
 - Principal components
 - Factor analysis
- Surprising observations
 - Mahalanobis distance
- Can we assume that the data follow a multivariate Normal distribution?

Section 1

Quantitative descriptive analysis

The data

The dataset contains 388 films and 15 variables.

Variables: Title, Year, Director, Country, ASL, and ten shot counts (RA, POV, INS, BCU, CU, MCU, MS, MLS, LS, VLS).

Median year 1937 (range 1912 to 2000), so the sample is concentrated in the 1920s and 1930s.

Missing values

Table 1: Variables with missing values

Variable	Missing
Year	2
BCU	1

Missingness is minimal: Year has 2 missing values and BCU has 1. Country has one blank level and one film with no country; Director has 16 blank entries.

Sample composition

Table 2: Number of films per country

Country	Films
USA	215
BRI	63
GER	55
FRA	37
ITA	5
SWE	5
RUS	3
OST	2
	1
AUS	1
HOL	1

The USA dominates (215 films, 55.4%), followed by Britain (63), Germany (55) and France (37); the remaining countries have 5 or fewer films

Univariate summaries

Table 3: Univariate numerical summaries

	Variable	Mean	Median	SD	IQR	Min	Max
ASL	ASL	11.79	7.7	50.59	5.00	0	1000
RA	RA	21.24	19.0	19.33	34.00	0	70
POV	POV	5.95	4.0	6.85	10.00	0	50
INS	INS	5.87	4.0	6.37	9.00	0	44
BCU	BCU	28.84	20.0	31.52	27.00	0	215
CU	CU	79.23	65.0	61.45	79.00	0	335
MCU	MCU	74.01	75.5	31.40	40.50	0	187
MS	MS	82.28	80.5	31.96	42.25	0	173
MLS	MLS	100.20	94.0	49.28	54.00	10	384
LS	LS	114.86	107.5	62.31	69.25	2	396
VLS	VLS	20.24	12.0	23.90	23.00	0	212

Reading the univariate table

ASL is strongly right-skewed: mean 11.79 s, median 7.70 s, SD 50.59 s, max 1000 s, min 0 s.

All count variables are overdispersed (variance greater than the mean), so median and IQR are more informative than mean and SD.

Most frequent shot type on average: LS (114.9), then MLS (100.2), MS (82.3), CU (79.2), MCU (74.0).

Least frequent: RA (21.2), VLS (20.2), POV (5.9), INS (5.9).

Overdispersion check

Table 4: Variance-to-mean ratio (VMR)

	Variable	Mean	Variance	VMR
ASL	ASL	11.8	2559.6	217.1
RA	RA	21.2	373.8	17.6
POV	POV	5.9	47.0	7.9
INS	INS	5.9	40.6	6.9
BCU	BCU	28.8	993.5	34.4
CU	CU	79.2	3776.0	47.7
MCU	MCU	74.0	985.9	13.3
MS	MS	82.3	1021.3	12.4
MLS	MLS	100.2	2428.9	24.2
LS	LS	114.9	3882.6	33.8
VLS	VLS	20.2	571.1	28.2

All VMR values are well above 1, confirming overdispersion.

Correlations (Spearman)

Table 5: Spearman correlation matrix

	ASL	RA	POV	INS	BCU	CU	MCU	MS	MLS	LS	VLS
ASL	1.00	-	-	-	-	-	-	0.27	0.46	0.16	0.01
		0.28	0.23	0.22	0.22	0.38	0.18				
RA	-	1.00	0.65	0.45	0.47	0.68	0.45	-	-	-	-
	0.28							0.20	0.52	0.52	0.38
POV	-	0.65	1.00	0.74	0.35	0.32	0.25	-	-	-	-
	0.23							0.15	0.24	0.12	0.19
INS	-	0.45	0.74	1.00	0.39	0.21	0.13	-	-	-	-
	0.22							0.17	0.20	0.06	0.10
BCU	-	0.47	0.35	0.39	1.00	0.60	0.21	-	-	-	-
	0.22							0.26	0.55	0.49	0.28
CU	-	0.68	0.32	0.21	0.60	1.00	0.53	-	-	-	-
	0.38							0.24	0.77	0.71	0.31
MCU	-	0.45	0.25	0.13	0.21	0.53	1.00	0.28	-	-	-
	0.18								0.42	0.63	0.22

Reading the correlations

Strong positive associations:

- POV and INS: 0.74
- RA and CU: 0.68
- BCU and CU: 0.60

Strong negative associations:

- CU and MLS: -0.77
- CU and LS: -0.71
- MCU and LS: -0.63

ASL is negatively correlated with RA (-0.28) and CU (-0.38), and positively with MLS (0.46).

This suggests a clear contrast between films built on close-ups and reverse angles and films built on longer, more distant shots.

Summaries by country

Table 6: Mean values by country ($n \geq 4$)

	Country	ASL	RA	CU	MLS	LS
3	BRI	8.8	28.0	89.2	92.6	105.5
4	FRA	9.6	10.5	45.7	99.9	148.4
5	GER	12.3	5.6	32.9	136.0	172.3
7	ITA	22.8	8.0	30.8	151.6	101.4
10	SWE	10.8	8.0	12.0	157.2	218.8
11	USA	12.8	26.0	96.9	91.1	93.9

Only the USA, Britain, Germany and France can be compared with any confidence.

Section 2

Surprising observations

Impossible values: $ASL = 0$

Table 7: Films with $ASL = 0$

	Title	Year	Director	ASL
139	Herrscher, Der	1937	Harlan, Veit	0
176	Kriemhilds Rache	1924	Lang, Fritz	0
327	Thomas Graals BÃœsta Film	1917	Stiller, Mauritz	0
364	Weekend	1968	Godard, Jean-Luc	0

An average shot length of zero seconds is impossible. These are almost certainly data errors or missing values coded as 0, and should be set to NA before further analysis.

A genuine extreme: Rope

Table 8: Film with the largest ASL

	Title	Year	Director	ASL
268	Rope	1948	Hitchcock, Alfred	1000

Rope (Hitchcock, 1948) has $ASL = 1000$ s.

This is not an error. Hitchcock deliberately shot Rope in a very small number of long continuous takes. An average of about 16.7 minutes per shot is consistent with that technique.

Lesson: an outlier can be scientifically meaningful, not just noise.

Extreme shot counts

Table 9: Extreme shot-count observations

	Title	BCU	CU	LS
65	Coronation Street (A)	81	254	7
70	Dallas - A Family Divided	96	260	10
72	Dallas - Who Done It?	54	322	21
75	Dark City	153	112	97
94	Driller Killer	154	122	34
97	Eastenders (?)	27	253	10
98	Eastenders (H)	76	268	12
135	Havsgamar	0	0	320
152	Insider, The	167	134	31
169	Karleken Segrar	42	0	301
NA	NA	NA	NA	NA
230	Neighbours	193	170	8
240	NYPD Blue	6	335	6
291	Silence of the Lambs, The	215	115	32

Blank country

Table 10: Film with a blank country

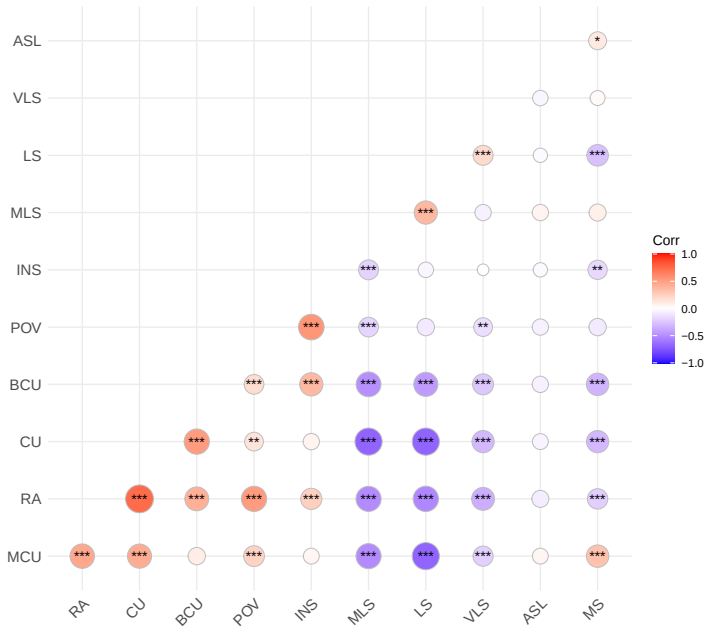
	Title	Year	Director	Country
73	D'Àmonit	1914		

One film has no country recorded and should be treated as Unknown or excluded.

Section 3

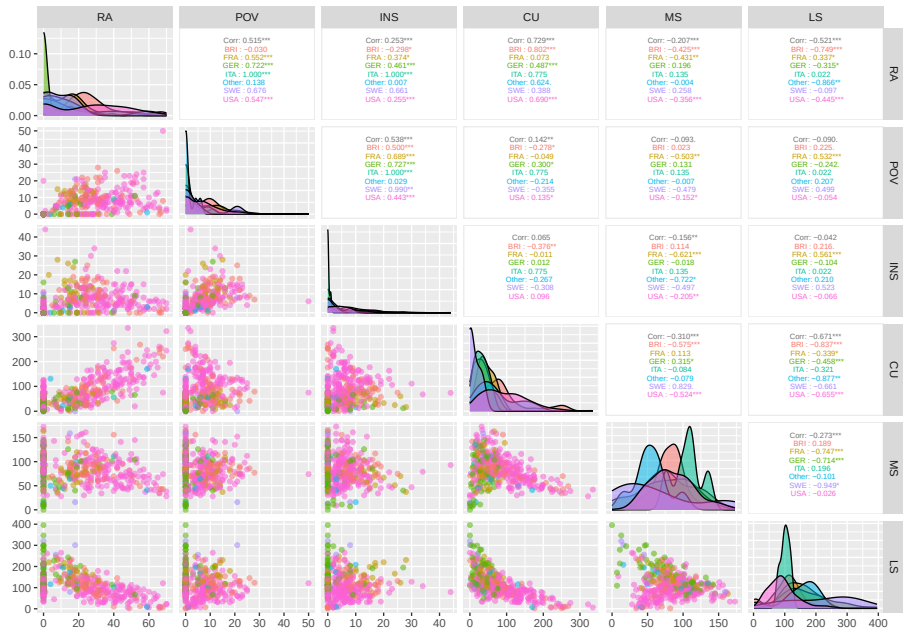
Graphical/qualitative analysis

Correlation matrix



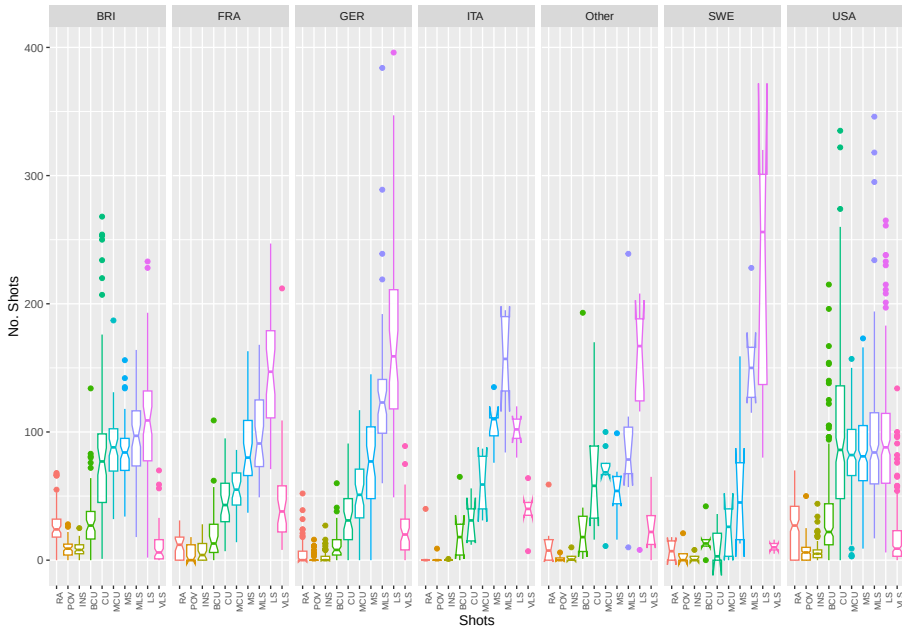
Average shot length does not depend on what shots are included. Medium shots have weak correlations with other shots, versatile shots that can be used in movies with a lot of close ups or long shots. Long shots have strong negative correlation with close up shots. Reverse angles have strong positive correlation with close up shots, shows conversations and character reactions.

Pairs



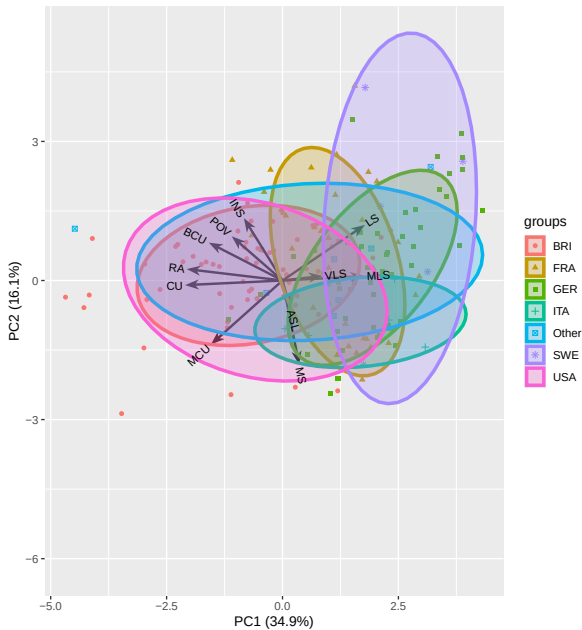
Displays shot types by country not including “medium/big/very” variants of close up/long shot. Movies from Italy have strong linear relationships between POV:reverse angle and inserted shot:reverse angle shots. Movies from Germany have very few reverse angle shots. Strong negative correlation between close up:long shot in all countries.

Boxplots



USA has similar mean shots for most shot types, likely because there are a lot more USA movies in dataset so have much greater variety of movies. All country groups except USA show a trend of higher shot counts as type moves from big close ups towards long shots, until very long shots which is consistently lower than big close up.

Principal Components

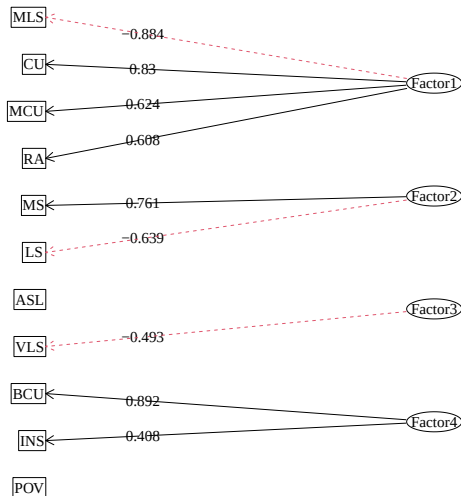


PC1: High MLS and VLS, low RA, CU, MCU PC2: High INS and LS, low MCU and MS

Sweden is furthest removed from the other country groups, and along with France varies much more along PC2 than PC1 Groups Britain, Italy, USA, and Other vary more across PC1 than PC2. Germany varies similarly across both.

Factor Analysis

Factor Analysis



Factor 1 contains 24% of variation, shows movies with high close up and reverse angle counts are associated with lower medium long shot counts. ASL and POV have high uniqueness and no strong associations to any factors. Factors do not capture variation in ASL and POV

Section 4

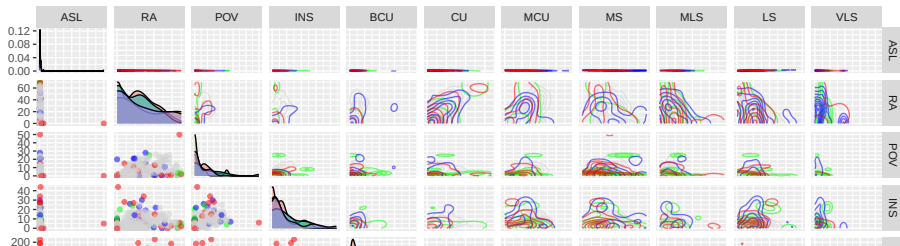
Surprising observations

Mahalanobis distance

##

##	Typical	Somewhat	Surprising	Very
##	325	11	23	27

##		Title	Year	dM	surprise
##	268	Rope	1948	380.93662	Very
##	211	Men of America	1932	88.25322	Very
##	366	West of Zanzibar	1928	83.85572	Very
##	364	Weekend	1968	77.68776	Very
##	53	Cape Forlorn	1930	66.08968	Very



Rope is by far the most surprising observation, directed for shots to be as long as could fit on film. 50 surprising/very surprising observations. Plot shows very surprising observations appear either at the extremes of number of shots, or with high densities around very few shots in interaction plots.

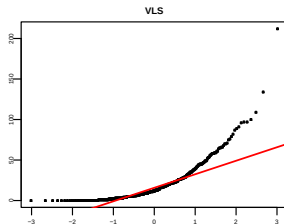
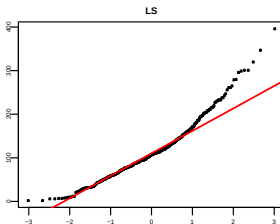
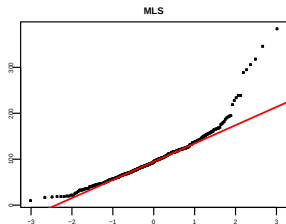
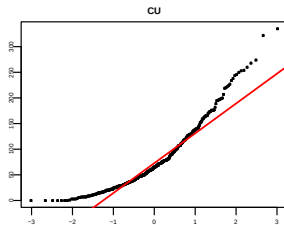
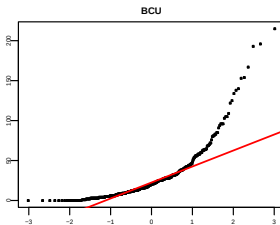
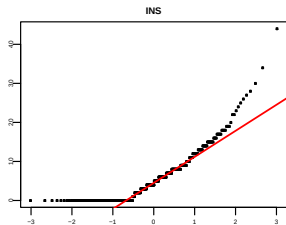
Can we assume the data follows a multivariate Normal distribution?

QQ-Norm plots

The QQ-Norm plots for each variable to check for normality

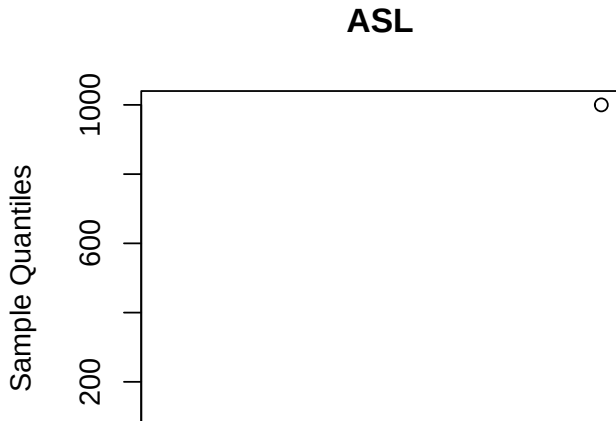
For the data to follow a multivariate normal distribution, each individual component must be normally distributed. We check this by visualising a QQ-Norm against QQ-line plot to visualize whether the data follows the line.

Q-Q plots: Variables that show they are not normally distributed



QQ-Normality - The variables that don't follow a normal distribution

- Based on this, we see that the variables INS, BCU, CU, MLS, LS, and VLS are not normally distributed.
- ALS also has an extreme outlier, producing a right-skew:



Anderson darling test of normality

We then perform an Anderson-Darling test of normality with a null hypothesis that the data does follow a normal distribution and the alternative hypothesis that the data does not follow a normal distribution. According to the results: - ASL, POV, INS, and VLS all show significant evidence against Normality, while RA, BCU, CU, MCU, MS, MLS, and LS do not provide evidence against normality. Due to the fact that multiple individuals fail a Normality check, we have evidence that the data are not multivariate normal.

##	ASL	RA	POV	INS	BCU	CU	MCU	MS	
##	0.0006	0.1904	0.1696	0.1834	0.2249	0.1646	0.6767	0.5011	0.0

Mahalanobis distance check for Multivariate Normality

The observed proportions of large Mahalanobis distances are greater than those expected under a multivariate normal distribution. These inconsistencies become particularly pronounced at the 95% and 99% cut-offs. The results therefore suggest that the movie data do not follow a multivariate normal distribution closely, with more extreme multivariate observations than would be expected under normality.

Based on the observed results, can we assume the data follow a multivariate normal distribution?

To conclude, we may assume that the data does not follow a multivariate normal distribution. Taken together, the marginal and multivariate diagnostics indicate that multivariate Normality is not a reasonable assumption for these data.

Are there groups in these data?

looking for groups

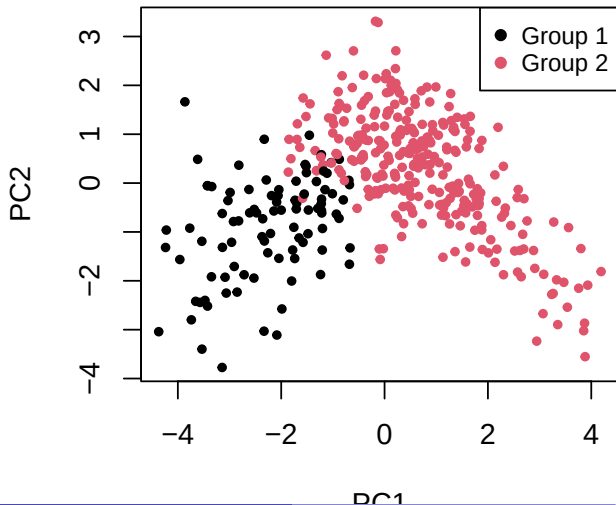
The seven shot-scale variables describe how shots are distributed from big close-ups to very long shots: BCU, CU, MCU, MS, MLS, LS, and VLS

Because the total number of these coded shots is approximately 500 for most films, we converted them to proportions before comparing films

We focus on shot scale rather than all 11 variables because: - the seven shot-scale variables directly describe the visual composition - RA, POV, and INS contain a lot of the same zero values - ASL contains an extreme value for Rope, which will affect the distance-based clustering strongly.

Checking for any distinct clusters

Shot-scale composition

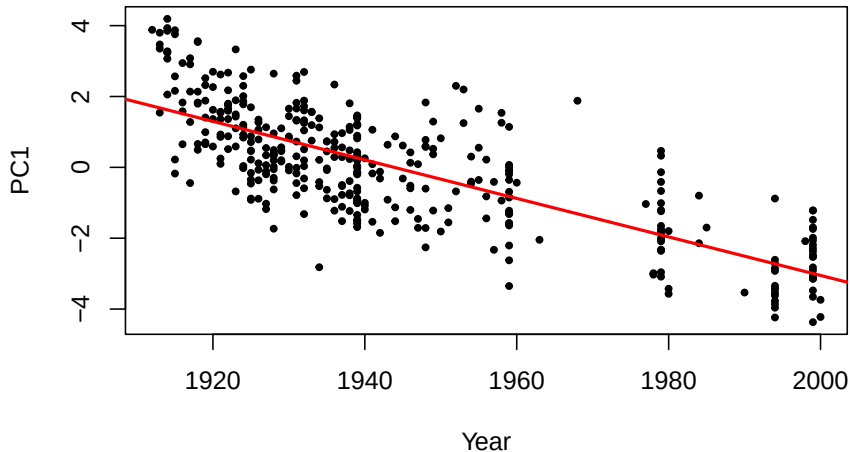


Discussion of shot-scale comparison

The Hierarchical clustering can divide the films into two groups but the PCA plot shows substantial overlap between them. This suggests that the films do not form two clearly separated populations. Instead, the clustering algorithm is dividing a more continuous pattern into two parts.

Shot scale comparison through time

Shot-scale composition through time



Taking a look at the smallest cluster

Examples from the distinctive early subset

Title

Year

CU

MLS

LS

388

Zweimal gelebt

1912

0

83

396

160

Are there groups?

No clear groups - The clustering algorithms can divide the films into groups, but the groups are not strongly separated. - The clearest structure:

- 1. a strong continuous change in shot-scale composition through time -
- 2. a small distinctive subset of films from approximately 1912-1916

Summary

Quantitative descriptive analysis:

- ASL and most counts are right-skewed and overdispersed, so use median and IQR.
- Strong correlations: CU with MLS (-0.77), POV with INS ($+0.74$).
- USA dominates; only USA, BRI, GER, FRA allow reliable country comparisons.

Qualitative analysis:

- Length of shots (ASL) is independent of types of shots.
- Movies with a lot of close ups and reverse shots include few long shots.
- Similar trends in shot types hold across all countries excluding USA

Surprising observations: - Four films with impossible $ASL = 0$, treat as missing. - Rope ($ASL = 1000$) is a real, meaningful extreme.

Can we assume the data follow a multivariate Normal distribution? - We have sufficient evidence to assume that the data does not follow a multivariate Normal distribution.